Department of Physics
National Institute of Technology Calicut

M.Sc. Physics Curriculum 2023

Module 2

Lattice Vibrations & Thermal Properties

Course Lecture Notes — Student Edition

PH7302E – Condensed Matter Physics (L–T–P–O: 3–1–0–8, C=4)

Lectures 8 – 16 • Course Outcome CO2

Where this module sits

Module 1 introduced the static, time-averaged crystal: lattice sites, the reciprocal lattice, and Bragg diffraction. Real atoms, however, are never at rest – they oscillate about those lattice sites. Module 2 develops the physics of these oscillations (*phonons*) and shows how they control the thermal and elastic behaviour of a solid. The results here (density of states, Bose statistics, the Debye model) are also the template used for the *electronic* density of states and statistics in Module 3.

Student Edition — Complete Lecture Notes, Derivations & Vector Diagrams

This student edition provides fully integrated theoretical derivations, step-by-step worked tutorial problems, and native vector diagrams for monoatomic and diatomic lattice vibrations, phonon density of states in 1D/2D/3D, classical, Einstein, and Debye heat capacities, and thermal conductivity mechanisms (Normal vs. Umklapp processes).

Contents

Notation Used in This Module **3**

1 Lecture 8: Harmonic Approximation and Monoatomic Linear Chain **4**
1.1 Introduction to Lattice Vibrations (8 min) . . . 4
1.2 Monoatomic Linear Chain (20 min) . . . 4
1.3 First Brillouin Zone (10 min) . . . 6
1.4 Density of States in 1D (7 min) . . . 6
1.5 Computational Illustration (5 min) . . . 6

2 Lecture 9: Diatomic Linear Chain and Optical Phonons **9**
2.1 Setup and Equations of Motion (12 min) . . . 9
2.2 Dispersion Relation Derivation (18 min) . . . 9
2.3 Analysis of Branches (12 min) . . . 10
2.4 Physical Interpretation (5 min) . . . 11
2.5 Computational Illustration (3 min) . . . 12

3 Lecture 10: Phonons – Quantization of Lattice Vibrations **13**
3.1 Classical to Quantum: Normal Modes (12 min) . . . 13
3.2 Phonon Creation and Annihilation (15 min) . . . 13
3.3 Bose–Einstein Statistics (12 min) . . . 14
3.4 Crystal Momentum Conservation (8 min) . . . 14
3.5 Computational Illustration (3 min) . . . 15

4 Lecture 11: Specific Heat – Classical and Einstein Models **16**
4.1 Classical Theory: Dulong–Petit Law (10 min) . . . 16
4.2 Einstein Model (20 min) . . . 16
4.3 Einstein Temperature and Physical Interpretation (10 min) . . . 17
4.4 Computational Illustration (10 min) . . . 17

5 Lecture 12: Debye Model of Specific Heat **19**
5.1 Motivation and Assumptions (8 min) . . . 19
5.2 Density of States in 3D (12 min) . . . 19
5.3 Debye Cutoff and Specific Heat (15 min) . . . 20
5.4 Limiting Behaviour (10 min) . . . 20
5.5 Computational Illustration (5 min) . . . 21

6 Lecture 13: Thermal Conductivity of Insulators **23**
6.1 Kinetic Theory Approach (12 min) . . . 23
6.2 Phonon Scattering Mechanisms (18 min) . . . 23
6.3 Debye–Peierls Theory (12 min) . . . 24
6.4 Computational Illustration (5 min) . . . 24

7 Lecture 14: Lattice Vibrations in 2D and 3D **26**
7.1 3D Monoatomic Lattice (15 min) . . . 26
7.2 3D Diatomic Lattice (12 min) . . . 26
7.3 Density of States in 2D and 3D (13 min) . . . 27
7.4 Computational Illustration (10 min) . . . 28

8 Lecture 15: Anharmonic Effects and Thermal Expansion **29**
8.1 Beyond the Harmonic Approximation (10 min) 29
8.2 Thermal Expansion (15 min) 30
8.3 Phonon–Phonon Interactions (15 min) 30
8.4 Grüneisen Parameter and Equation of State (7 min) 31
8.5 Computational Illustration (3 min) 31

9 Lecture 16: Tutorial & Problem Solving – Lattice Vibrations **33**
Problem Set 33
Computational Challenge 35

Module 2 Summary Sheet **36**

Further Reading **37**

Notation Used in This Module

Symbol	Meaning
a	lattice spacing (1D) / lattice constant
u_n, v_n	displacement of the n-th atom (two sublattices in a diatomic chain)
C	nearest-neighbour force (spring) constant
k	phonon wavevector
$\omega(k)$	phonon angular frequency (dispersion relation)
$v_g = d\omega/dk$	group velocity; v_s = sound (phase) velocity at $k \to 0$
$g(\omega)$	density of states (modes per unit frequency)
$n_B(\omega)$	Bose–Einstein occupation number of a phonon mode
Θ_E, Θ_D	Einstein temperature, Debye temperature
ω_D, k_D	Debye cutoff frequency and wavevector
κ	thermal conductivity
ℓ	phonon mean free path
γ	Grüneisen parameter

1 Lecture 8: Harmonic Approximation and Monoatomic Linear Chain

Duration: 50 minutes

Learning Objectives

- Understand the harmonic approximation in lattice dynamics
- Derive the dispersion relation for a monoatomic 1D chain
- Distinguish between acoustic and optical modes

† A short history

The lattice-vibration problem was set up by Born and von Kármán in 1912–13, who first wrote down the periodic boundary conditions and the $3N$ normal modes of a finite crystal. The dispersion relation $\omega(k) = 2\sqrt{C/M}\,|\sin(ka/2)|$ that we are about to derive is, in essence, what they obtained. The interpretation of these modes as *phonons* (quantised lattice waves) was firmly established only after 1932, following Dirac's work on field quantisation. Einstein's 1907 paper on specific heat used a much cruder (single-frequency) model that we will see in Lecture 11.

1.1 Introduction to Lattice Vibrations (8 min)

Physical Picture

Atoms in a crystal are not stationary at lattice sites but vibrate about their equilibrium positions. These vibrations are responsible for:

- Thermal properties (specific heat, thermal expansion, thermal conductivity)
- Elastic properties
- Electrical resistivity (electron–phonon scattering)
- Superconductivity

Harmonic Approximation. For small displacements u_n from equilibrium, the potential energy is expanded as

$$U = U_0 + \sum_n \frac{1}{2} C\,(u_n - u_{n+1})^2 + \cdots$$

where C is the force constant (spring constant between nearest neighbours). Higher-order (anharmonic) terms are neglected – we return to them in Lecture 15.

Key assumption

Only nearest-neighbour interactions matter.

! Why only nearest neighbours?

For the monatomic chain, the nearest-neighbour approximation is mostly pedagogical – it gives the simplest correct answer. In a real 3D solid, second and further neighbour shells can shift the dispersion curve by 10–30%. Modern *ab initio* lattice dynamics fits the force-constant matrix $\Phi_{\alpha\beta}$ to density-functional theory forces, often including 4–6 neighbour shells.

1.2 Monoatomic Linear Chain (20 min)

Setup.

- Atoms of mass M arranged in a 1D lattice with spacing a
- Displacement of the n-th atom: u_n

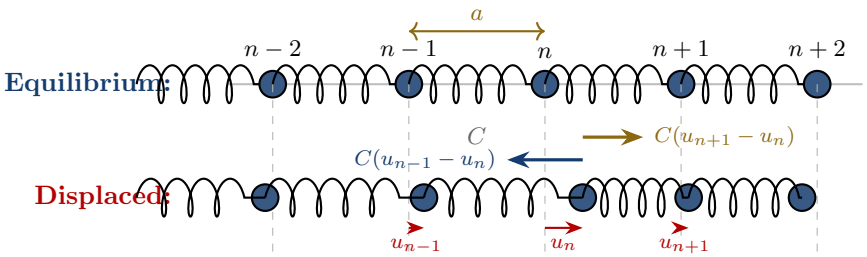


Figure 1: One-dimensional monoatomic harmonic chain. (Top) Equilibrium positions with lattice constant a. (Bottom) Dynamic state showing instantaneous atomic displacements u_n and net restoring forces exerted by adjacent nearest-neighbour springs of force constant C.

- Only nearest-neighbour springs, force constant C

Equation of motion.

$$M\frac{d^2u_n}{dt^2} = C(u_{n+1} - u_n) - C(u_n - u_{n-1}) = C(u_{n+1} + u_{n-1} - 2u_n)$$

Trial solution (plane wave).

$$u_n = A\,e^{i(kna-\omega t)}$$

where k is the wavevector and ω the angular frequency.

Substitution.

$$-M\omega^2 = C\left(e^{ika} + e^{-ika} - 2\right) = 2C(\cos ka - 1) = -4C\sin^2\left(\tfrac{ka}{2}\right)$$

Dispersion Relation

$$\omega(k) = 2\sqrt{\frac{C}{M}}\left|\sin\left(\frac{ka}{2}\right)\right|$$

Key features.

1. **Periodic in k:** $\omega(k + 2\pi/a) = \omega(k) \Rightarrow$ only $k \in [-\pi/a, \pi/a]$ (first Brillouin zone) is physically distinct.
2. **Linear at small k:** $\omega \approx v_s|k|$, with $v_s = a\sqrt{C/M}$ the sound velocity.
3. **Maximum at zone boundary:** $\omega_{\max} = 2\sqrt{C/M}$ at $k = \pm\pi/a$.
4. **Group velocity:** $v_g = d\omega/dk = a\sqrt{C/M}\,\cos(ka/2) \to 0$ at the zone boundary.

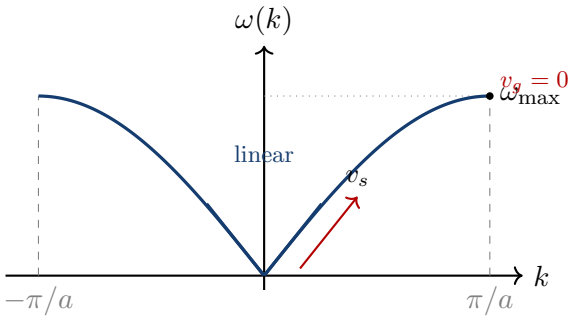


Figure 2: Dispersion $\omega(k)$ for the monoatomic chain. Linear (sound) near $k = 0$ with slope v_s; saturates at $\omega_{\max} = 2\sqrt{C/M}$ at the zone boundary $k = \pm\pi/a$, where the group velocity vanishes.

! "Group velocity = 0" does not mean the mode is a standing wave

At the zone boundary, two counter-propagating plane waves of equal amplitude form a standing wave, but a normal mode of the chain can be *any* phase combination of them. The vanishing of v_g simply means that this particular mode does not transport energy – useful for thermal-resistance arguments later.

1.3 First Brillouin Zone (10 min)

Definition. The first Brillouin zone is the Wigner–Seitz cell of the reciprocal lattice. It contains all unique values of k for which the dispersion relation is not repeated.

For the 1D chain:

- Reciprocal lattice vector: $G = 2\pi/a$
- First BZ: $k \in [-\pi/a, \pi/a]$

Physical significance.

- Modes with k and $k + G$ are physically identical (same atomic displacements).
- The number of allowed k-values in the first BZ equals the number of unit cells N.
- With periodic boundary conditions: $k = \dfrac{2\pi n}{Na}$, $n = 0, \pm1, \pm2, \ldots, \pm N/2$.

Recall from Module 1

This is exactly the reciprocal-lattice / Wigner–Seitz construction you used for X-ray diffraction, now applied to *vibrational* wavevectors instead of scattering wavevectors.

⊛ How we measure this dispersion: inelastic neutron scattering

Neutrons are the workhorse for phonon dispersion measurements because their kinetic energy (∼meV–eV) matches phonon energies, and they couple directly to atomic nuclei (not to electron density). A monochromatic neutron beam of wavevector $\mathbf{k}_i$ is scattered into $\mathbf{k}_f$, with energy loss $\hbar\omega = \frac{\hbar^2}{2m_n}(k_i^2 - k_f^2)$ and momentum transfer $\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i$. Conservation modulo a reciprocal lattice vector gives a phonon dispersion point $(\mathbf{Q}, \omega)$. Pioneered at Brookhaven and Chalk River in the 1960s; standard textbook experiment.

1.4 Density of States in 1D (7 min)

Definition. The density of states $g(\omega)$ gives the number of modes per unit frequency interval.

For the 1D monoatomic chain:

$$g(\omega)\, d\omega = \frac{L}{\pi}\, dk = \frac{L}{\pi}\frac{dk}{d\omega}\, d\omega, \qquad g(\omega) = \frac{L}{\pi v_g} = \frac{L}{\pi a\sqrt{C/M}\,\cos(ka/2)}$$

Using $\cos(ka/2) = \sqrt{1 - \sin^2(ka/2)} = \sqrt{1 - (\omega/\omega_{\max})^2}$:

1D Density of States

$$g(\omega) = \frac{2L}{\pi\sqrt{\omega_{\max}^2 - \omega^2}}$$

This diverges at $\omega = \omega_{\max}$ – a **Van Hove singularity**, arising because $v_g \to 0$ there (flat top of the dispersion curve).

1.5 Computational Illustration (5 min)

The dispersion relation and group velocity of the monoatomic chain can be visualised directly from the closed-form expressions derived above:

$$\omega(k) = 2\sqrt{C/M}\,|\sin(ka/2)|, \qquad v_g(k) = a\sqrt{C/M}\,\cos(ka/2).$$

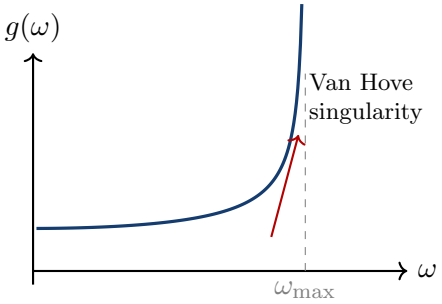
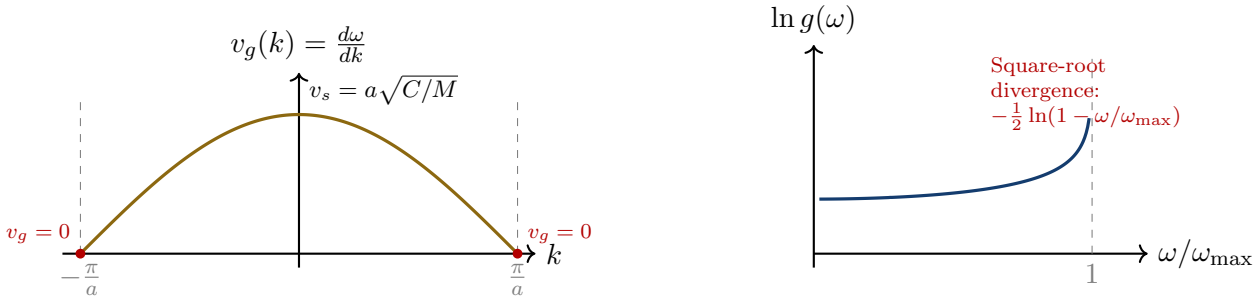


Figure 3: 1D monoatomic chain density of states. The square-root divergence at $\omega_{\max}$ is the simplest example of a Van Hove singularity, arising from the vanishing group velocity at the zone boundary.



(a) Phonon Group Velocity $v_g(k)$ **(b) Logarithmic DOS** $\ln g(\omega)$

Figure 4: (a) Phonon group velocity $v_g(k) = v_s \cos(ka/2)$, demonstrating the acoustic limit ($v_g = v_s$ at $k = 0$) and zero velocity at the Brillouin zone edge. (b) 1D density of states on a logarithmic scale, highlighting the integrable Van Hove singularity at $\omega = \omega_{\max}$.

Plotting both across the first Brillouin zone $k \in [-\pi/a, \pi/a]$ makes the zero-group-velocity condition at the zone boundary especially clear – this is a good in-class or take-home numerical exercise (e.g. using NumPy/Matplotlib).

Python: dispersion and group velocity of a monoatomic chain

```python
import numpy as np
import matplotlib.pyplot as plt

# --- Parameters (SI) ---
M = 1.66e-27 * 40       # 40 amu -> kg
C = 50.0                # N/m
a = 3.0e-10             # m

# Derived quantities
omega_max = 2.0*np.sqrt(C/M)          # rad/s
v_s       = a*np.sqrt(C/M)            # sound velocity, m/s

# Wavevector sweep across the first Brillouin zone
k = np.linspace(-np.pi/a, np.pi/a, 1001)
omega = omega_max * np.abs(np.sin(k*a/2.0))
v_g   = a*np.sqrt(C/M) * np.cos(k*a/2.0)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9, 3.5))

ax1.plot(k*a, omega)
ax1.set_xlabel(r'$ka$')
ax1.set_ylabel(r'$\omega(k)$ (rad/s)')
ax1.set_title('Dispersion')
```

```
ax1.axvline(np.pi, color='gray', ls=':')
ax1.text(np.pi, omega_max*0.95, r'$\pi$', ha='center')

ax2.plot(k*a, v_g)
ax2.set_xlabel(r'$ka$')
ax2.set_ylabel(r'$v_g(k)$ (m/s)')
ax2.set_title('Group velocity')
ax2.axhline(0, color='k', lw=0.5)
ax2.axvline(np.pi, color='gray', ls=':')

plt.tight_layout()
plt.show()

# Sanity check: print the speed of sound
print(f"v_s = {v_s:.3e} m/s,  omega_max = {omega_max:.3e} rad/s")
```

★ Take-home message of Lecture 8

A 1D monatomic chain has N modes in the first BZ, with dispersion $\omega(k) = 2\sqrt{C/M}\,|\sin(ka/2)|$. The mode is acoustic: $\omega \to 0$ as $k \to 0$. The 1D DOS diverges as $1/\sqrt{\omega_{\max}^2 - \omega^2}$ at the zone edge.

Tutorial / Practice Problems

1. Show that the sound velocity in a monoatomic chain is $v_s = a\sqrt{C/M}$.
2. Calculate the density of states at $\omega = \omega_{\max}/2$ for a chain of 1000 atoms.
3. What happens to the group velocity at the zone boundary? Explain physically.

2 Lecture 9: Diatomic Linear Chain and Optical Phonons

Duration: 50 minutes

Learning Objectives

- Derive the dispersion relation for a diatomic chain
- Distinguish acoustic and optical branches
- Understand the physical nature of optical modes

† Why the diatomic chain is worth knowing

The diatomic chain is the simplest model that exhibits the two essential features of any crystal with more than one atom per unit cell: two branches (acoustic, optical) and a *forbidden frequency gap* at the zone boundary. The same physics controls the existence of electronic band gaps in semiconductors – the mathematical structure (a 2×2 secular equation) is identical, with "mass" replaced by "on-site energy" and "spring constant" replaced by "hopping amplitude."

2.1 Setup and Equations of Motion (12 min)

Setup.

- Two types of atoms, masses M and m ($M > m$)
- Alternating arrangement with spacing $a/2$ between neighbours
- Lattice constant a (repeat distance)
- Same force constant C for all nearest-neighbour bonds

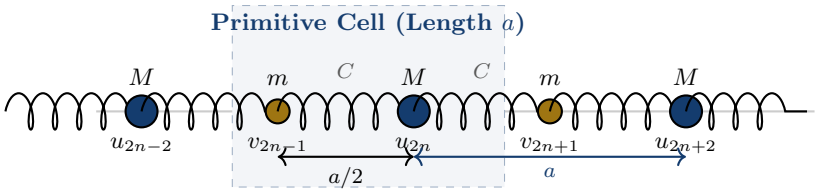


Figure 5: One-dimensional diatomic lattice chain with alternating masses $M > m$ and identical spring constants C. The primitive unit cell has length a and contains two basis atoms with equilibrium spacing $a/2$.

Equations of motion.

Even sites (mass M):

$$M\frac{d^2u_{2n}}{dt^2} = C(v_{2n+1} + v_{2n-1} - 2u_{2n})$$

Odd sites (mass m):

$$m\frac{d^2v_{2n+1}}{dt^2} = C(u_{2n+2} + u_{2n} - 2v_{2n+1})$$

Trial solutions.

$$u_{2n} = A\,e^{i(nka-\omega t)}, \qquad v_{2n+1} = B\,e^{i(nka+ka/2-\omega t)}$$

2.2 Dispersion Relation Derivation (18 min)

Substituting the trial solutions:

$$-M\omega^2 A = C\left(2B\cos\tfrac{ka}{2} - 2A\right), \qquad -m\omega^2 B = C\left(2A\cos\tfrac{ka}{2} - 2B\right)$$

Rearranged into matrix form:

$$(2C - M\omega^2)A - 2C\cos\left(\tfrac{ka}{2}\right)B = 0, \qquad -2C\cos\left(\tfrac{ka}{2}\right)A + (2C - m\omega^2)B = 0$$

Secular equation (determinant = 0):

$$\begin{vmatrix} 2C - M\omega^2 & -2C\cos(ka/2) \\ -2C\cos(ka/2) & 2C - m\omega^2 \end{vmatrix} = 0$$

$$(2C - M\omega^2)(2C - m\omega^2) - 4C^2\cos^2\left(\tfrac{ka}{2}\right) = 0$$

Expanded (filled-in step). Let $x \equiv \omega^2$. Multiplying out:

$$4C^2 - 2C(M+m)x + Mmx^2 - 4C^2\cos^2\left(\tfrac{ka}{2}\right) = 0.$$

Using $\cos^2(\theta) = \frac{1}{2}(1 + \cos 2\theta)$ and $1 - \cos 2\theta = 2\sin^2\theta$:

$$4C^2\left[1 - \cos^2\left(\tfrac{ka}{2}\right)\right] = 4C^2\sin^2\left(\tfrac{ka}{2}\right),$$

so the secular equation simplifies to

Quadratic in ω^2

$$Mm\,\omega^4 - 2C(M+m)\,\omega^2 + 4C^2\sin^2\left(\tfrac{ka}{2}\right) = 0$$

The quadratic formula in ω^2 gives

Dispersion Relation – Two Branches

$$\omega_{\pm}^2 = C\left(\frac{1}{M} + \frac{1}{m}\right) \pm C\sqrt{\left(\frac{1}{M} + \frac{1}{m}\right)^2 - \frac{4\sin^2(ka/2)}{Mm}}$$

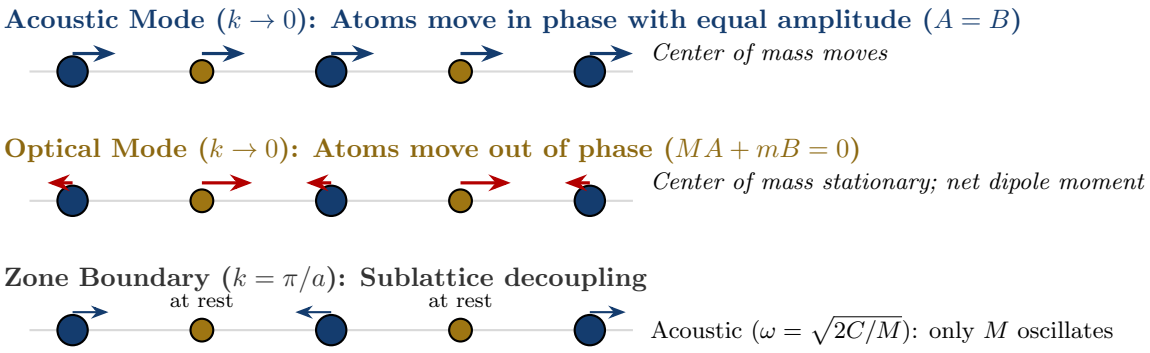


Figure 6: Physical displacement configurations of the diatomic chain. In the acoustic branch at $k \to 0$, basis atoms move together in phase. In the optical branch at $k \to 0$, opposite charges oscillate against each other, producing an oscillating electric dipole moment. At the zone boundary $k = \pi/a$, the two sublattices decouple completely.

2.3 Analysis of Branches (12 min)

Acoustic branch (ω_-).

- At $k = 0$: $\omega_- = 0$, $A = B$ (atoms move in phase).
- At $k = \pi/a$: $\omega_- = \sqrt{2C/M}$ (for $M > m$); $A/B = 0$ (heavy atoms move, light atoms stationary).

- Linear at small k: $\omega_- \approx v_s|k|$, $v_s = a\sqrt{C/(2(M+m))}$.
- Called "acoustic" because at long wavelength these are ordinary sound waves.

Optical branch (ω_+).

- At $k = 0$: $\omega_+ = \sqrt{2C(1/M + 1/m)}$, $A/B = -m/M$ (atoms move out of phase).
- At $k = \pi/a$: $\omega_+ = \sqrt{2C/m}$, $B = 0$ (light atoms move, heavy atoms stationary).
- Called "optical" because in ionic crystals these modes couple to infrared light (the oscillating electric field drives oppositely-charged ions in opposite directions).

Key observation

There is a **frequency gap** between $\sqrt{2C/M}$ and $\sqrt{2C/m}$ at $k = \pi/a$. No propagating modes exist in this gap.

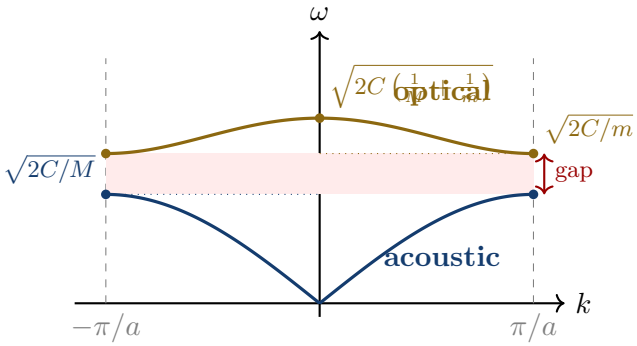


Figure 7: Schematic diatomic-chain dispersion. The shaded band between $\sqrt{2C/M}$ and $\sqrt{2C/m}$ is the forbidden frequency gap. Real systems have this gap renormalised by the actual force-constant tensor, but the topology is generic.

2.4 Physical Interpretation (5 min)

Acoustic mode, $k \approx 0$

Both sublattices move together – a macroscopic elastic deformation.

Optical mode, $k \approx 0$

Sublattices move against each other – no centre-of-mass motion, but internal strain. In polar crystals this creates an oscillating dipole moment that couples to electromagnetic radiation.

Zone boundary, $k = \pi/a$

One sublattice is stationary while the other vibrates; which one depends on the branch.

⊛ Direct observation: Raman and infrared spectroscopy

The $k = 0$ optical mode is the one that couples to light. In a polar (ionic) crystal like NaCl or GaAs, this mode is *infrared active* – a long-wavelength EM wave drives the two oppositely-charged sublattices against each other. In non-polar crystals (Si, Ge, diamond) the $k = 0$ optical mode can be *Raman active*: the oscillating strain modulates the electronic polarisability, scattering light inelastically with a frequency shift equal to the optical phonon frequency. Raman and IR are thus complementary, and the LO–TO splitting (Lec 14) controls which modes are seen in which experiment.

2.5 Computational Illustration (3 min)

Python: plot the two branches of the diatomic chain

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters (arbitrary units: M=2, m=1, C=1, a=1)
M, m, C, a = 2.0, 1.0, 1.0, 1.0

k = np.linspace(-np.pi/a, np.pi/a, 1001)
s = np.sin(k*a/2.0)

# Closed-form omega_+^2 and omega_-^2
inv = (1.0/M + 1.0/m)
disc = np.sqrt(inv**2 - 4.0*s**2/(M*m))
omega_p = np.sqrt(C*(inv + disc))
omega_m = np.sqrt(C*(inv - disc))

# Sanity checks
print(f"omega_- at k=0      = {omega_m[500]:.4f}  (should be 0)")
print(f"omega_- at k=pi/a   = {omega_m[-1]:.4f}  (should be sqrt(2C/M) = {np.sqrt(2*C/M):.4f})")
print(f"omega_+ at k=0      = {omega_p[500]:.4f}  (should be sqrt(2C/M+2C/m) = {np.sqrt(2*C*(1/M+1/m)):.4f})")
print(f"omega_+ at k=pi/a   = {omega_p[-1]:.4f}  (should be sqrt(2C/m) = {np.sqrt(2*C/m):.4f})")

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(k*a, omega_p, color='tab:orange', label=r'optical $\omega_+$')
ax.plot(k*a, omega_m, color='tab:blue',   label=r'acoustic $\omega_-$')
ax.set_xlabel(r'$ka$')
ax.set_ylabel(r'$\omega_\pm(k)$')
ax.axvspan(0, np.pi, color='red', alpha=0.06)
ax.legend()
plt.tight_layout()
plt.show()
```

! "Acoustic" and "optical" names are not about which atoms move

The naming refers to *what the mode couples to*, not who is moving. Acoustic modes are the ones that, at long wavelength, are sound – they couple to macroscopic strain. Optical modes couple to electromagnetic radiation (in polar materials). A heavy atom moving alone at the zone boundary is still an acoustic mode if the dispersion branch is the lower one; the labels are fixed by the branch, not the instantaneous displacement pattern.

Tutorial / Practice Problems

1. Show that for $M = m$ the diatomic chain reduces to the monoatomic case with lattice constant $a/2$.
2. Calculate the frequency gap for a diatomic chain with $M = 40$ amu, $m = 20$ amu, $C = 50$ N/m.
3. Explain why the optical branch is flat near $k = 0$ in real 3D crystals.

3 Lecture 10: Phonons – Quantization of Lattice Vibrations

Duration: 50 minutes

Learning Objectives

- Understand phonons as quantum quasiparticles
- Derive phonon energy and momentum
- Apply Bose–Einstein statistics to phonons

† Phonons: a 20th-century name for an old idea

The word "phonon" was coined by Yakov Frenkel in 1932, drawing an explicit analogy with the photon. The underlying physics is just the second-quantised harmonic oscillator: any normal mode of the lattice, treated quantum mechanically, is a bosonic field with energy quanta $\hbar\omega$. The Bose–Einstein distribution that we will use was developed by Bose (1924) and Einstein (1924–25) for photons; its application to phonons is a one-line replacement of the dispersion.

3.1 Classical to Quantum: Normal Modes (12 min)

Classical picture. A crystal with N atoms has $3N$ degrees of freedom. In the harmonic approximation the Hamiltonian diagonalises into $3N$ independent harmonic oscillators (normal modes).

Quantum picture. Each normal mode of frequency ω is quantised like a harmonic oscillator:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad n = 0, 1, 2, \ldots$$

Definition – Phonon

A phonon is the quantum of lattice vibrational energy. It is a **quasiparticle** with

- Energy $E = \hbar\omega$
- (Crystal) momentum $\mathbf{p} = \hbar\mathbf{k}$
- Spin 0 (boson)

Important distinction

Phonons are *not* real particles carrying physical momentum. $\hbar\mathbf{k}$ is *crystal momentum* – conserved only modulo a reciprocal lattice vector. Phonons are collective excitations of the entire lattice.

3.2 Phonon Creation and Annihilation (15 min)

Ladder operators. For each mode $(\mathbf{k}, s)$, s labelling polarisation:

$$a^\dagger_{\mathbf{k}s}\,|n_{\mathbf{k}s}\rangle = \sqrt{n_{\mathbf{k}s}+1}\;|n_{\mathbf{k}s}+1\rangle\,, \qquad a_{\mathbf{k}s}\,|n_{\mathbf{k}s}\rangle = \sqrt{n_{\mathbf{k}s}}\;|n_{\mathbf{k}s}-1\rangle$$

Hamiltonian.

$$H = \sum_{\mathbf{k},s} \hbar\omega_{\mathbf{k}s}\left(a^\dagger_{\mathbf{k}s}a_{\mathbf{k}s} + \frac{1}{2}\right)$$

Number operator. $n_{\mathbf{k}s} = a^\dagger_{\mathbf{k}s}a_{\mathbf{k}s}$ gives the number of phonons in mode $(\mathbf{k}, s)$.

Physical interpretation

- $a^\dagger_{\mathbf{k}s}$ creates a phonon; $a_{\mathbf{k}s}$ annihilates one.
- The ground state ($T = 0$) has **zero-point energy** $\sum \hbar\omega/2$.

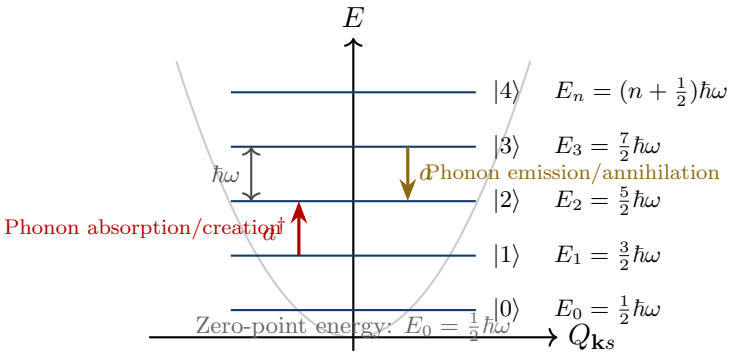


Figure 8: Harmonic oscillator energy ladder for a single normal mode $(\mathbf{k}, s)$. The creation operator $a^\dagger$ adds a phonon of energy $\hbar\omega$, while the annihilation operator a removes one. The vacuum ground state has persistent zero-point energy $\frac{1}{2}\hbar\omega$.

! "Crystal momentum is conserved modulo G"

This is the single most-violated rule in solid-state homework. When two phonons collide, $\mathbf{k}_1 + \mathbf{k}_2$ need not equal $\mathbf{k}_3$; it only has to equal $\mathbf{k}_3$ plus *some* reciprocal lattice vector. The cases $\mathbf{G} = 0$ and $\mathbf{G} \neq 0$ are called *normal* and *Umklapp* processes, respectively. The latter are essential for thermal resistance (Lec 13) and for electrical resistivity at high T.

3.3 Bose–Einstein Statistics (12 min)

Phonons are bosons (integer spin, no Pauli exclusion). The average occupation number of a phonon mode at temperature T is the **Planck** (Bose–Einstein) distribution:

Bose–Einstein Occupation

$$\langle n_{\mathbf{k}s} \rangle = \frac{1}{e^{\hbar\omega_{\mathbf{k}s}/k_B T} - 1} = n_B(\omega_{\mathbf{k}s})$$

Limiting cases.

- High T ($k_B T \gg \hbar\omega$): $n_B \approx k_B T/\hbar\omega \gg 1$ (classical equipartition).
- Low T ($k_B T \ll \hbar\omega$): $n_B \approx e^{-\hbar\omega/k_B T} \ll 1$ (quantum regime).

Total energy.

$$U = \sum_{\mathbf{k},s} \hbar\omega_{\mathbf{k}s} \left(\langle n_{\mathbf{k}s} \rangle + \frac{1}{2} \right)$$

3.4 Crystal Momentum Conservation (8 min)

Umklapp processes. In phonon–phonon or phonon–electron interactions, crystal momentum is conserved *modulo* a reciprocal lattice vector:

$$\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{G}$$

- $\mathbf{G} = 0$: **Normal process** – total crystal momentum conserved.
- $\mathbf{G} \neq 0$: **Umklapp process** – crystal momentum changes by $\mathbf{G}$.

Physical significance

Umklapp processes are crucial for thermal resistance (they can reverse the direction of heat flow) and for electrical resistivity at high temperature.

⊛ Phonon population measured by inelastic scattering

The phonon occupation number $n_B(\omega, T)$ is directly probed by Raman and neutron scattering: the intensity ratio of the Stokes (+1 phonon created) to anti-Stokes (−1 phonon annihilated) lines is

$$\frac{I_{\text{Stokes}}}{I_{\text{anti-Stokes}}} = \frac{n_B(\omega)+1}{n_B(\omega)} = e^{\hbar\omega/k_BT}.$$

Measuring this ratio as a function of T is one of the cleanest ways to verify the Bose–Einstein distribution.

3.5 Computational Illustration (3 min)

Python: Bose--Einstein distribution and the Stokes/anti-Stokes ratio

```python
import numpy as np
import matplotlib.pyplot as plt

kB = 1.380649e-23
hbar = 1.054571817e-34
# phonon frequency (rad/s) -- mid-range optical mode
omega = 1.0e13

def n_B(T):
    return 1.0/(np.exp(hbar*omega/(kB*T)) - 1.0)

# Temperature sweep
T_vals = np.linspace(5, 300, 200)
nB     = n_B(T_vals)

# Stokes / anti-Stokes ratio -- direct check of BE statistics
ratio = (nB + 1)/nB
expected = np.exp(hbar*omega/(kB*T_vals))

fig, ax = plt.subplots(1, 2, figsize=(9,3.3))
ax[0].semilogy(T_vals, nB)
ax[0].set_xlabel('T (K)'); ax[0].set_ylabel(r'$\langle n_B(\omega)\rangle$')
ax[0].set_title('Bose--Einstein occupation vs T')

ax[1].plot(T_vals, ratio, label='(n+1)/n')
ax[1].plot(T_vals, expected, '--', label=r'$\exp(\hbar\omega/k_BT)$')
ax[1].set_xlabel('T (K)'); ax[1].set_ylabel('Stokes/anti-Stokes')
ax[1].legend()
plt.tight_layout(); plt.show()
```

Tutorial / Practice Problems

1. Calculate the zero-point energy per atom for a Debye solid with $\Theta_D = 300$ K.
2. At what temperature does the average phonon occupation number equal 1 for a mode with $\omega = 10^{13}$ rad/s?
3. Explain why phonons obey Bose statistics rather than Fermi statistics.

4 Lecture 11: Specific Heat – Classical and Einstein Models

Duration: 50 minutes

Learning Objectives

- Derive the Dulong–Petit law
- Understand the Einstein model and its predictions
- Compare with experimental data

† A puzzle from 1819

Dulong and Petit noticed in 1819 that the heat capacity per mole of nearly every solid is about 25 J/K at room temperature. A century later, low-temperature measurements (Nernst, 1910) showed that C_V falls to zero as $T \to 0$ – a behaviour the equipartition theorem cannot explain. Einstein's 1907 paper was the first quantum-mechanical fix; Debye's 1912 refinement introduced the linear acoustic dispersion we will see in Lec 12.

4.1 Classical Theory: Dulong–Petit Law (10 min)

Equipartition theorem. Each quadratic degree of freedom contributes $\frac{1}{2}k_BT$ to the energy. For a solid of N atoms, each atom has 3 vibrational degrees of freedom (kinetic + potential = 2 per direction):

$$U = 3N \times 2 \times \frac{1}{2}k_BT = 3Nk_BT$$

Specific Heat

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = 3Nk_B = 3R \quad \text{(per mole)}$$

Dulong–Petit law: $C_V \approx 25$ J/mol·K for all solids at high T.
Failure at low T. Experiments show $C_V \to 0$ as $T \to 0$, contradicting the classical prediction.

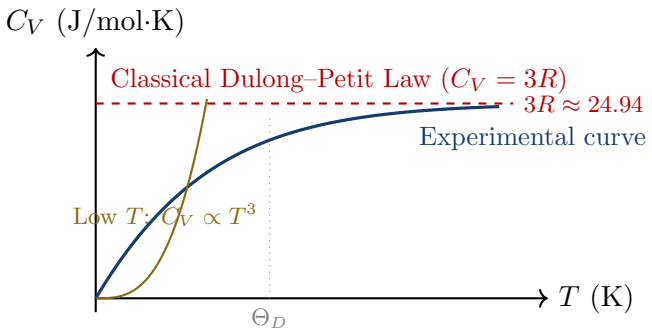


Figure 9: Molar specific heat $C_V(T)$ of a typical non-magnetic dielectric crystal. At high temperatures ($T \gg \Theta_D$), C_V approaches the classical Dulong–Petit limit $3R$. At low temperatures ($T \ll \Theta_D$), quantum freeze-out suppresses C_V with a characteristic T^3 power law.

4.2 Einstein Model (20 min)

Assumption. All $3N$ oscillators share the *same* frequency ω_E (the Einstein frequency).
Energy.

$$U = 3N\,\frac{\hbar\omega_E}{e^{\hbar\omega_E/k_BT} - 1} + \frac{3N\hbar\omega_E}{2} \quad \text{(zero-point term dropped for } C_V\text{)}$$

Einstein Specific Heat

$$C_V = 3Nk_B \left(\frac{\Theta_E}{T}\right)^2 \frac{e^{\Theta_E/T}}{\left(e^{\Theta_E/T}-1\right)^2}, \qquad \Theta_E \equiv \frac{\hbar\omega_E}{k_B}$$

Limiting behaviour.

- High T ($T \gg \Theta_E$): $C_V \to 3Nk_B$ (Dulong–Petit, correct!)
- Low T ($T \ll \Theta_E$): $C_V \approx 3Nk_B(\Theta_E/T)^2 e^{-\Theta_E/T}$ – exponential decay

Comparison with experiment.

- High T: correct.
- Low T: experiment shows $C_V \propto T^3$, but Einstein predicts a faster (exponential) decay. The discrepancy arises because Einstein ignores the low-frequency *acoustic* modes.

4.3 Einstein Temperature and Physical Interpretation (10 min)

Typical values.

Material	Θ_E (K)
Diamond	≈ 1320 (very stiff bonds)
Copper	≈ 240
Lead	≈ 68 (soft, heavy)

Physical meaning

Θ_E characterises lattice stiffness: high Θ_E means stiff bonds, light atoms, high vibrational frequencies. It is treated as a fitting parameter and works reasonably well for optical modes, but fails for acoustic modes.

! Θ_E and Θ_D are not the same number

For the same material, Θ_E and Θ_D are not interchangeable. The Einstein temperature is set by an optical-mode frequency; the Debye temperature is a particular average over the acoustic dispersion. Numerically, Θ_D is typically 70–80% of Θ_E – a rule of thumb, not an identity.

4.4 Computational Illustration (10 min)

Python: Einstein specific heat and the Dulong--Petit limit

```python
import numpy as np
import matplotlib.pyplot as plt
kB = 1.380649e-23
NA = 6.02214076e23
R  = NA*kB

def Cv_Einstein(T, ThetaE):
    x = ThetaE/T
    # numerical guard against T=0
    x = np.where(x > 500, 500, x)
    return 3*R * x**2 * np.exp(x) / (np.exp(x) - 1.0)**2

T = np.linspace(1, 1000, 1000)
for ThetaE in (68, 240, 1320):   # Pb, Cu, diamond
    plt.plot(T, Cv_Einstein(T, ThetaE), label=fr'$\Theta_E={ThetaE}$~K')

plt.axhline(3*R, color='k', ls=':', label='Dulong--Petit 3R')
plt.xscale('log')
plt.xlabel('T (K)'); plt.ylabel(r'$C_V$ (J/mol.K)')
plt.ylim(0, 30); plt.legend()
```

```
plt.tight_layout(); plt.show()
```

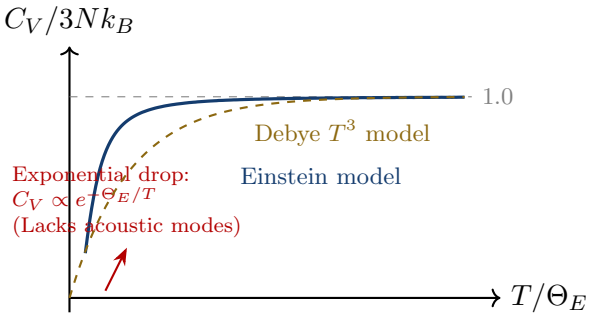


Figure 10: Dimensionless heat capacity $C_V/3Nk_B$ according to the Einstein model as a function of reduced temperature T/Θ_E. While capturing the high-temperature saturation at $3Nk_B$, it drops off exponentially at low temperatures ($e^{-\Theta_E/T}$), failing to predict the correct experimental T^3 acoustic dependence.

Tutorial / Practice Problems

1. Calculate the Einstein temperature for a solid with $\omega_E = 10^{13}$ rad/s.
2. Show that the Einstein specific heat approaches $3Nk_B$ at high T.
3. At what fraction of Θ_E does C_V reach 90% of the Dulong–Petit value?

5 Lecture 12: Debye Model of Specific Heat

Duration: 50 minutes

Learning Objectives

- Derive the Debye model using a linear dispersion approximation
- Understand the Debye temperature and cutoff frequency
- Reproduce the T^3 law at low temperature

† Peter Debye, 1912

Debye's 1912 paper was a single tidy move: replace the unrealistic single-frequency assumption with a realistic linear acoustic dispersion $\omega = v_s k$, and let the cutoff be set by counting modes. The T^3 law at low T drops out of the Debye integral essentially automatically. It is one of the cleanest successes of early quantum theory.

5.1 Motivation and Assumptions (8 min)

Einstein model failure. At low T, only low-frequency (acoustic) modes are excited, but the Einstein model treats all modes as high-frequency optical modes.

Debye's insight. At low T, only long-wavelength acoustic modes matter, and these obey a **linear dispersion**: $\omega = v_s k$.

Assumptions.

1. Linear dispersion for *all* modes: $\omega = v_s k$.
2. Isotropic solid with an average sound velocity v_s.
3. A cutoff frequency ω_D (Debye frequency) such that the total number of modes equals $3N$.

! What the Debye model gets wrong

The Debye model assumes $\omega = v_s k$ up to the cutoff. Real solids have a *flat* optical-like region near the zone boundary where v_g goes to zero. This means the true high-T C_V is slightly below $3R$ and the approach to the limit is not exactly the Dulong–Petit curve. The model is, however, excellent at low T (where only the linear regime matters) and qualitatively correct at all temperatures.

5.2 Density of States in 3D (12 min)

For a continuous isotropic medium, the number of modes with $|\mathbf{k}| < k$:

$$N(k) = \frac{V}{(2\pi)^3} \times \frac{4\pi k^3}{3} \times 3 = \frac{V k^3}{2\pi^2}$$

(the factor of 3 counts polarisations: 1 longitudinal + 2 transverse).

Density of states in k-space:

$$g(k)\,dk = \frac{V k^2}{2\pi^2}\,dk$$

Converting to frequency ($\omega = v_s k$):

3D Debye Density of States

$$g(\omega)\,d\omega = \frac{V\omega^2}{2\pi^2 v_s^3}\,d\omega$$

With different velocities for longitudinal and transverse modes:

$$\frac{3}{v_s^3} = \frac{1}{v_l^3} + \frac{2}{v_t^3}$$

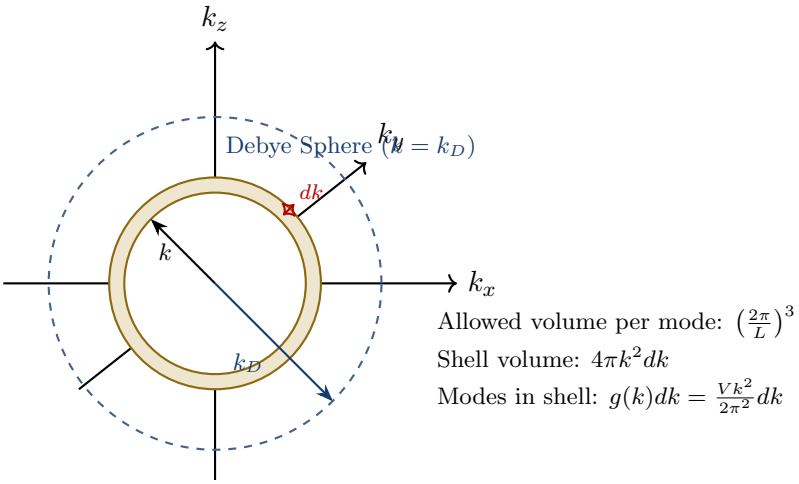


Figure 11: Mode counting in three-dimensional **k**-space. The differential spherical shell between radius k and $k + dk$ contains $4\pi k^2 dk/(2\pi/L)^3$ spatial modes. The Debye sphere of radius k_D encompasses exactly $3N$ normal vibrational modes.

5.3 Debye Cutoff and Specific Heat (15 min)

Debye frequency ω_D: set the total number of modes equal to $3N$:

$$\int_0^{\omega_D} g(\omega)\, d\omega = 3N \implies \frac{V}{2\pi^2 v_s^3}\int_0^{\omega_D} \omega^2\, d\omega = 3N \implies \frac{V\omega_D^3}{6\pi^2 v_s^3} = 3N$$

Debye wavevector:

$$k_D = \left(\frac{6\pi^2 N}{V}\right)^{1/3}$$

Debye temperature:

$$\Theta_D = \frac{\hbar\omega_D}{k_B} = \frac{\hbar v_s k_D}{k_B}$$

Specific heat derivation:

$$U = \int_0^{\omega_D} \hbar\omega\, \langle n(\omega)\rangle\, g(\omega)\, d\omega$$

Debye Specific Heat

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = 9Nk_B\left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x-1)^2}\, dx, \qquad x \equiv \frac{\hbar\omega}{k_B T}$$

5.4 Limiting Behaviour (10 min)

High T **(**$T \gg \Theta_D$**):** expand $e^x \approx 1 + x$:

$$C_V \approx 9Nk_B\left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} x^2\, dx = 9Nk_B\left(\frac{T}{\Theta_D}\right)^3 \cdot \frac{1}{3}\left(\frac{\Theta_D}{T}\right)^3 = 3Nk_B$$

$\Rightarrow$ Dulong–Petit law recovered.

Low T ($T \ll \Theta_D$): the upper limit $\Theta_D/T \to \infty$, and

$$\int_0^\infty \frac{x^4 e^x}{(e^x-1)^2}\,dx = \frac{4\pi^4}{15}$$

Debye T^3 Law

$$C_V = \frac{12\pi^4}{5} N k_B \left(\frac{T}{\Theta_D}\right)^3 \propto T^3$$

This matches experimental data extremely well for non-magnetic insulators. **Where the $12\pi^4/5$ comes from:** it is $3 \times 4\pi^4/15$ – the factor 3 is the three acoustic polarisations, $4\pi^4/15$ is the standard Bose–Einstein integral $\int_0^\infty x^4 e^x/(e^x-1)^2\,dx$.

⊛ Why low-T C_V is the cleanest test of any phonon model

The Debye T^3 law is so robust that deviations from it are diagnostic. For example, in glasses the specific heat goes as $C_V \propto T$ (two-level systems) below 1 K – a famous signature of amorphous solids. In ferroelectrics near T_C a "critical" contribution appears on top of the Debye T^3. Plotting C_V/T^3 vs. T is therefore a standard first-look diagnostic in any low-T specific-heat experiment.

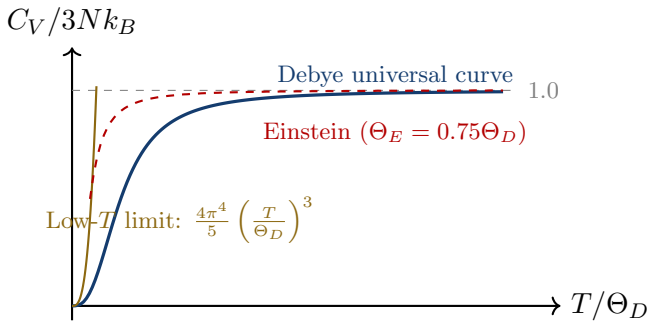


Figure 12: Universal Debye specific heat $C_V/3Nk_B$ as a function of reduced temperature T/Θ_D. At low temperatures ($T/\Theta_D < 0.1$), the curve follows the exact T^3 tangent. The Einstein curve with matched median frequency drops significantly faster at low temperatures.

Summary – Three Models Compared

	Classical	Einstein	Debye
Mode frequencies	– (equipartition)	all $= \omega_E$	continuum, $\omega = v_s k$, cutoff ω_D
High-T limit	$3Nk_B$	$3Nk_B$	$3Nk_B$
Low-T limit	$3Nk_B$ (wrong)	$\propto e^{-\Theta_E/T}$	$\propto T^3$ (correct)

5.5 Computational Illustration (5 min)

Python: Debye specific heat, low-T T^3, and Einstein overlay

```python
import numpy as np
from scipy.integrate import quad
import matplotlib.pyplot as plt

kB = 1.380649e-23
```

```python
R  = 8.314462618

def integrand(x):
    return x**4 * np.exp(x) / (np.exp(x) - 1.0)**2

def Cv_Debye(T, ThetaD):
    upper = ThetaD/T if T > 0 else 700
    upper = min(upper, 700)   # numerical cutoff
    val, _ = quad(integrand, 0, upper)
    return 9*R * (T/ThetaD)**3 * val

def Cv_Einstein(T, ThetaE):
    x = ThetaE/T
    return 3*R * x**2 * np.exp(x) / (np.exp(x) - 1.0)**2

T = np.linspace(1, 600, 600)
ThetaD = 320.0  # K
Cv_D = np.array([Cv_Debye(t, ThetaD) for t in T])
Cv_E = Cv_Einstein(T, ThetaD)   # using ThetaD as ThetaE for the overlay

plt.plot(T, Cv_D, label='Debye')
plt.plot(T, Cv_E, '--', label=fr'Einstein, $\Theta_E=\Theta_D$')
plt.axhline(3*R, color='k', ls=':', label='Dulong--Petit')
plt.xlabel('T (K)'); plt.ylabel(r'$C_V$ (J/mol.K)')
plt.legend(); plt.tight_layout(); plt.show()
```

Tutorial / Practice Problems

1. Calculate the Debye temperature for copper given $v_l = 4760$ m/s, $v_t = 2325$ m/s, and atomic density $n = 8.5 \times 10^{28}$ m^{-3}.
2. Show that the Debye model reduces to Dulong–Petit at high T.
3. At $T = 10$ K, calculate the specific heat of Al ($\Theta_D = 394$ K) using the Debye model.

6 Lecture 13: Thermal Conductivity of Insulators

Duration: 50 minutes

Learning Objectives

- Understand the kinetic theory of thermal conductivity
- Derive the phonon thermal conductivity formula
- Explain the role of phonon scattering mechanisms

⊛ What $\kappa(T)$ actually looks like

The thermal conductivity of pure dielectric crystals (e.g. NaF, sapphire, diamond) is one of the most striking non-monotonic curves in solid-state physics. At the lowest T, $\kappa \propto T^3$ (specific heat rising, mean free path set by the sample size – Casimir regime). It peaks at a few percent of Θ_D, then falls as $1/T$ at high T (Umklapp-dominated). The peak value can exceed 10^4 W/m·K in isotopically pure diamond at low T – the highest thermal conductivity of any known solid.

6.1 Kinetic Theory Approach (12 min)

Thermal conductivity from kinetic theory:

$$\kappa = \frac{1}{3} C_V v \ell$$

where C_V is the specific heat per unit volume, v the particle velocity, and ℓ the mean free path.

For phonons:

Phonon Thermal Conductivity

$$\kappa = \frac{1}{3} C_V v_s \ell$$

Physical Picture

Heat flows because phonons with higher energy (from the hot side) diffuse toward the cold side. The mean free path ℓ is limited by scattering processes.

6.2 Phonon Scattering Mechanisms (18 min)

1. Phonon–phonon scattering (Umklapp processes). At high T, phonon density is high ⇒ frequent collisions. Only Umklapp processes (which change total crystal momentum) contribute to thermal resistance.

Temperature dependence:

- High T: $\ell \propto 1/T$ (phonon density $\propto T$) ⇒ $\kappa \propto 1/T$ (since C_V is constant at high T).

2. Boundary scattering. At low T, the phonon mean free path is limited by sample dimensions:

$$\ell \approx L \text{ (sample size)}, \qquad \kappa = \frac{1}{3} C_V v_s L \propto T^3$$

3. Impurity/defect scattering. Point defects, isotopes and dislocations scatter phonons. At low T:

$$\ell \propto 1/\omega^4 \propto 1/T^4 \qquad \text{(Rayleigh scattering, } \sigma \propto \omega^4\text{)}$$

4. Electron–phonon scattering. In metals, electrons also scatter and are scattered by phonons.

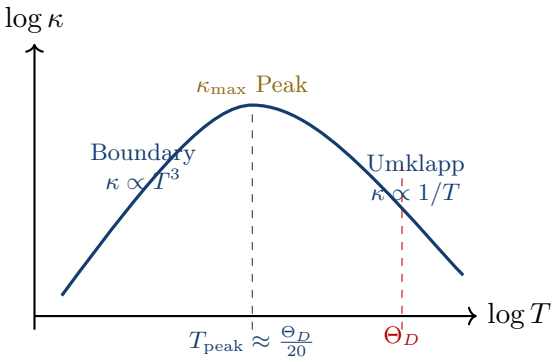


Figure 13: Schematic $\kappa(T)$ for a pure dielectric crystal on log–log axes. Three regimes: T^3 (boundary-limited), peak near $T \approx \Theta_D/20$, then $1/T$ (Umklapp-limited).

6.3 Debye–Peierls Theory (12 min)

Peierls' contribution. Normal phonon processes ($\mathbf{G} = 0$) do not contribute to thermal resistance because total crystal momentum is conserved. Only Umklapp processes create resistance.

Umklapp condition. At least one phonon must have $k > \pi/2a$ (half the zone boundary). The probability of such high-k phonons:

$$P \propto e^{-\Theta_D/2T}$$

Resulting Thermal Conductivity

$\kappa \propto e^{\Theta_D/2T}$ (intermediate T); $\kappa \propto 1/T$ (high T); $\kappa \propto T^3$ (low T, boundary limited)

6.4 Computational Illustration (5 min)

Python: Matthiessen's-rule model of $\kappa(T)$

```python
import numpy as np
import matplotlib.pyplot as plt

# Simple model: combine boundary (Casimir) and Umklapp
# kappa(T) = (1/3) * Cv(T) * v_s * l_eff,  l_eff^-1 = l_b^-1 + l_U^-1

def Cv_Debye_per_vol(T, ThetaD, rho):
    # approximate low-T Cv ~ (12 pi^4 / 5) * (N/V) * kB * (T/ThetaD)^3
    n = rho / (40*1.66e-27)   # for 40 amu, e.g. Ca-like
    a = 12*np.pi**4/5 * n * 1.38e-23 * (T/ThetaD)**3
    return np.where(T < ThetaD/2, a, 3*n*1.38e-23)   # plateau at Dulong-Petit

v_s    = 4000.0
rho    = 3000.0
ThetaD= 300.0
L      = 1e-3   # 1 mm sample

T = np.linspace(1, 500, 500)
Cv  = Cv_Debye_per_vol(T, ThetaD, rho)
l_b = L*np.ones_like(T)
l_U = (1e-2) * (ThetaD/T)        # crude Umklapp MFP, ~ 10 microns at T=ThetaD
l_eff = 1.0/(1.0/l_b + 1.0/l_U)
kappa = (1.0/3.0) * Cv * v_s * l_eff

plt.loglog(T, kappa)
```

```
plt.xlabel('T (K)'); plt.ylabel(r'$\kappa$ (W/m.K)')
plt.title('Matthiessen model: boundary + Umklapp')
plt.tight_layout(); plt.show()
```

Tutorial / Practice Problems

1. Estimate the phonon mean free path in diamond at room temperature given $\kappa = 2000$ W/m·K, $v_s = 12000$ m/s, and $C_V = 1.8 \times 10^6$ J/m^3·K.
2. Why does the thermal conductivity of amorphous solids not show T^3 behaviour at low T?
3. Explain the role of isotope scattering in reducing thermal conductivity.

7 Lecture 14: Lattice Vibrations in 2D and 3D

Duration: 50 minutes

Learning Objectives

- Extend lattice dynamics to higher dimensions
- Understand polarisation and the number of branches
- Calculate the density of states in 2D and 3D

7.1 3D Monoatomic Lattice (15 min)

Setup. N atoms in a 3D Bravais lattice, each with 3 degrees of freedom $\Rightarrow 3N$ normal modes.

Dispersion relation. For each $\mathbf{k}$ in the first Brillouin zone there are **3 branches**:

- 1 longitudinal acoustic (LA) branch
- 2 transverse acoustic (TA) branches

Equations of motion:

$$M\frac{d^2u_\alpha(\mathbf{R})}{dt^2} = -\sum_{\mathbf{R}'}\sum_{\beta}\Phi_{\alpha\beta}(\mathbf{R}-\mathbf{R}')\,u_\beta(\mathbf{R}')$$

where $\Phi_{\alpha\beta}$ is the force-constant matrix.

Dynamical matrix:

$$D_{\alpha\beta}(\mathbf{k}) = \frac{1}{M}\sum_{\mathbf{R}}\Phi_{\alpha\beta}(\mathbf{R})\,e^{-i\mathbf{k}\cdot\mathbf{R}}$$

The eigenvalues of $D_{\alpha\beta}(\mathbf{k})$ give $\omega^2(\mathbf{k})$ for the three branches.

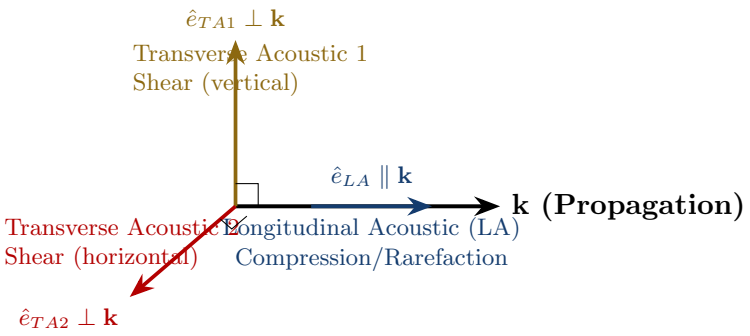


Figure 14: Polarization eigenvectors for acoustic vibrational waves in a three-dimensional continuum. For each wavevector $\mathbf{k}$, there are three mutually orthogonal acoustic modes: one longitudinal wave ($\hat{e}_{LA} \parallel \mathbf{k}$) and two transverse waves ($\hat{e}_{TA1}, \hat{e}_{TA2} \perp \mathbf{k}$).

7.2 3D Diatomic Lattice (12 min)

Setup. Basis of 2 atoms per primitive cell $\Rightarrow 6N$ normal modes ($3 \times 2N$ atoms).

Branches.

- 3 acoustic branches (1 LA + 2 TA)
- 3 optical branches (1 LO + 2 TO)

LO–TO splitting. In polar crystals (ionic bonds), the longitudinal optical (LO) and transverse optical (TO) modes differ in frequency at $\mathbf{k} = 0$ due to long-range electrostatic forces:

Lyddane–Sachs–Teller Relation

$$\omega_{LO}^2 - \omega_{TO}^2 = \frac{ne^2}{\mu\epsilon_0\epsilon_\infty}$$

where μ is the reduced mass and ϵ_∞ the high-frequency dielectric constant.

! Lyddane–Sachs–Teller is only at $\mathbf{k} = 0$ and only for polar crystals

The LST relation $\omega_{LO}^2/\omega_{TO}^2 = \epsilon_0/\epsilon_\infty$ (in the simpler form) holds *only* at $\mathbf{k} = 0$ and *only* when the long-range Coulomb interaction between ions lifts the LO–TO degeneracy. It does *not* apply in non-polar crystals (Si, Ge, diamond) where LO and TO are degenerate at $\mathbf{k} = 0$.

7.3 Density of States in 2D and 3D (13 min)

General formula:

$$g(\omega) = \sum_s \int_{BZ} \frac{d^d k}{(2\pi)^d}\, \delta(\omega - \omega_s(\mathbf{k}))$$

3D:

$$g(\omega) = \frac{V}{2\pi^2} \sum_s \frac{k^2}{|v_{gs}|}, \qquad g(\omega) \propto \omega^2 \quad \text{(isotropic linear dispersion)}$$

2D:

$$g(\omega) = \frac{A}{2\pi} \sum_s \frac{k}{|v_{gs}|}, \qquad g(\omega) \propto \omega \quad \text{(isotropic linear dispersion)}$$

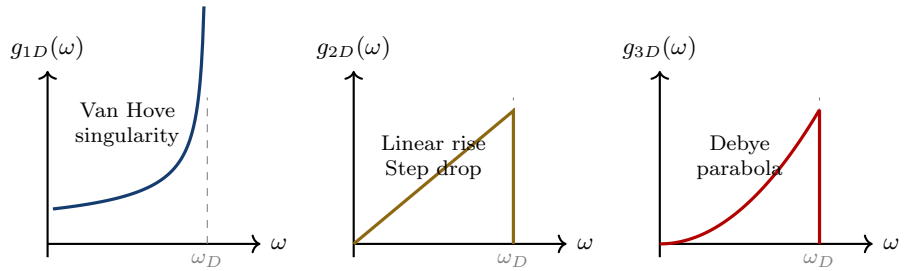
1D (recap):

$$g(\omega) = \frac{L}{\pi} \sum_s \frac{1}{|v_{gs}|}$$

Dimensional pattern

$g(\omega) \propto \omega^{d-1}$ for a linear, isotropic dispersion in d dimensions – a useful mnemonic linking the 1D, 2D and 3D results above.

Van Hove singularities. These occur at critical points where $v_g = d\omega/dk = 0$ (stationary points of the dispersion). $g(\omega)$ diverges or has discontinuous derivatives there.



1D: $g(\omega) \propto \omega^0/v_g$ **2D:** $g(\omega) \propto \omega$ **3D:** $g(\omega) \propto \omega^2$

Figure 15: Comparison of phonon density of states $g(\omega) \propto \omega^{d-1}$ for isotropic linear dispersion in $d = 1, 2, 3$ dimensions. (Left) 1D chain exhibiting inverse square-root Van Hove singularity. (Center) 2D membrane exhibiting linear frequency dependence and discontinuous step at cutoff. (Right) 3D Debye continuum exhibiting parabolic ω^2 dependence.

7.4 Computational Illustration (10 min)

For a 2D square lattice with nearest-neighbour springs, the (simplified isotropic) dispersion is

$$\omega(\mathbf{k}) = \sqrt{\frac{2C}{M}\,(2 - \cos k_x a - \cos k_y a)}.$$

Plotting ω along the high-symmetry path $\Gamma \to X \to M \to \Gamma$ produces a standard "phonon band structure" plot; plotting the full 3D surface $\omega(k_x, k_y)$ over the square Brillouin zone shows the saddle point at X that produces a Van Hove singularity in $g(\omega)$.

Python: 2D square-lattice dispersion along Γ--X--M--Γ

```python
import numpy as np
import matplotlib.pyplot as plt

# 2D square lattice dispersion (C = M = a = 1, arbitrary units)
def omega2D(kx, ky):
    return 2.0*(2.0 - np.cos(kx) - np.cos(ky))

# high-symmetry path:  G(0,0) -> X(pi,0) -> M(pi,pi) -> G(0,0)
def kpath(t):
    # t in [0,1]; 0..0.25 -> G->X, 0.25..0.5 -> X->M, 0.5..1 -> M->G
    if t < 0.25:
        s = t/0.25
        return np.array([np.pi*s, 0.0])
    elif t < 0.5:
        s = (t-0.25)/0.25
        return np.array([np.pi, np.pi*s])
    else:
        s = (t-0.5)/0.5
        return np.array([np.pi*(1-s), np.pi*(1-s)])

t = np.linspace(0, 1, 400)
ks = np.array([kpath(ti) for ti in t])
om = np.sqrt(omega2D(ks[:,0], ks[:,1]))

plt.plot(t, om)
xticks = [0, 0.25, 0.5, 1.0]
xlabels = [r'$\Gamma$', 'X', 'M', r'$\Gamma$']
plt.xticks(xticks, xlabels)
plt.ylabel(r'$\omega(\mathbf k)$')
plt.title('2D square-lattice phonon dispersion')
plt.tight_layout(); plt.show()
```

Tutorial / Practice Problems

1. How many acoustic and optical branches does a 3D crystal with 4 atoms per primitive cell have?
2. Derive the DOS for a 2D Debye model and show $C_V \propto T^2$ at low T.
3. Explain why graphene has a linear dispersion near the K points (Dirac cones).

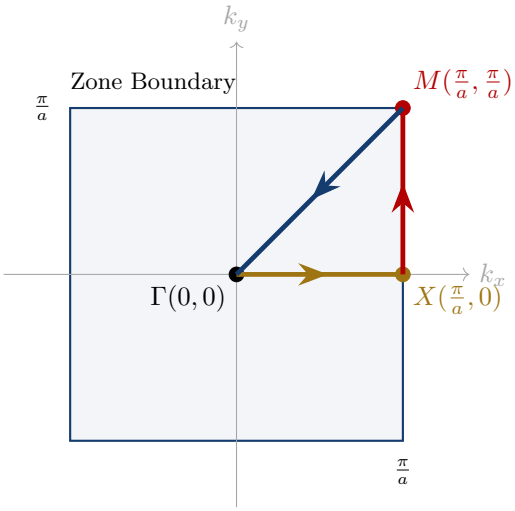


Figure 16: The first Brillouin zone of a 2D square lattice with lattice constant a. The irreducible high-symmetry path $\Gamma(0,0) \to X(\frac{\pi}{a},0) \to M(\frac{\pi}{a},\frac{\pi}{a}) \to \Gamma(0,0)$ spans all independent vibrational wavevectors.

8 Lecture 15: Anharmonic Effects and Thermal Expansion

Duration: 50 minutes

Learning Objectives

- Understand the origin of thermal expansion
- Derive the Grüneisen parameter
- Connect anharmonicity to thermal conductivity

† Eigenvalue Grüneisen, 1926

Eduard Grüneisen introduced the dimensionless parameter γ in 1926 to relate thermal expansion to specific heat. The relation $\beta = \gamma C_V/(3B)$ is so robust that γ is tabulated for essentially every solid. Negative thermal expansion (Lec 15 problem) was conclusively observed in ZrW_2O_8 by Mary et al. in 1996 – a 70-year-old prediction.

8.1 Beyond the Harmonic Approximation (10 min)

Harmonic approximation limitation. In the harmonic model, phonons do not interact. This predicts (all wrongly):

- Infinite thermal conductivity (no phonon–phonon scattering)
- No thermal expansion (symmetric potential)
- Elastic constants independent of temperature

All three are violated in reality due to **anharmonicity** (higher-order terms in the potential).

Anharmonic potential:

$$U = U_0 + \frac{1}{2}C_2x^2 + \frac{1}{3!}C_3x^3 + \frac{1}{4!}C_4x^4 + \cdots$$

The cubic term C_3x^3 is the leading anharmonic correction.

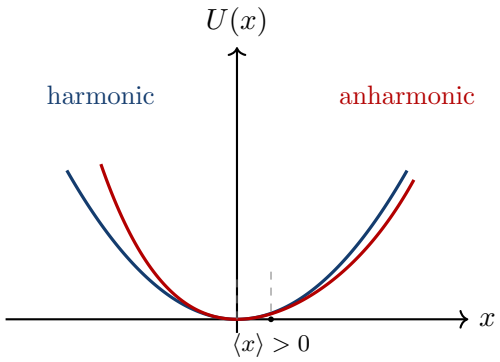


Figure 17: Harmonic (symmetric) vs. anharmonic (asymmetric) potential. The anharmonic potential well has a flatter right wall ($x > 0$) due to the negative cubic term ($-cx^3$) and a steeper left wall ($x < 0$), causing the thermal equilibrium displacement to expand toward positive separations ($\langle x \rangle > 0$).

8.2 Thermal Expansion (15 min)

Physical origin

In a symmetric (harmonic) potential the average position of an oscillator sits at the minimum regardless of its energy. In an asymmetric (anharmonic) potential, higher-energy states have their average position shifted to larger separations.

Grüneisen model. The Grüneisen parameter γ relates thermal expansion to phonon frequencies:

Grüneisen Parameter

$$\gamma = -\frac{d \ln \omega}{d \ln V}$$

For a mode with frequency ω:

$$\omega = \omega_0 - \gamma \omega_0 \frac{\Delta V}{V}$$

Thermal expansion coefficient:

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P = \frac{\gamma C_V}{3B}$$

where B is the bulk modulus.

Key results.

- Low T: $\beta \propto T^3$ (since $C_V \propto T^3$).
- High T: $\beta \approx$ constant (since $C_V \approx 3Nk_B$).

8.3 Phonon–Phonon Interactions (15 min)

Three-phonon processes. The cubic anharmonic term allows three-phonon interactions,

$$\omega_1 + \omega_2 = \omega_3$$

with momentum conservation (normal or Umklapp).

Normal processes ($\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3$):

- Do not contribute to thermal resistance.
- Redistribute energy among modes but conserve total crystal momentum.

Umklapp processes ($\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{G}$):

- Change total crystal momentum.
- Dominant source of thermal resistance at high T.
- Require high-energy phonons ($k > \pi/2a$).
- Freeze out exponentially at low T: $\tau_U^{-1} \propto Te^{-\Theta_D/3T}$.

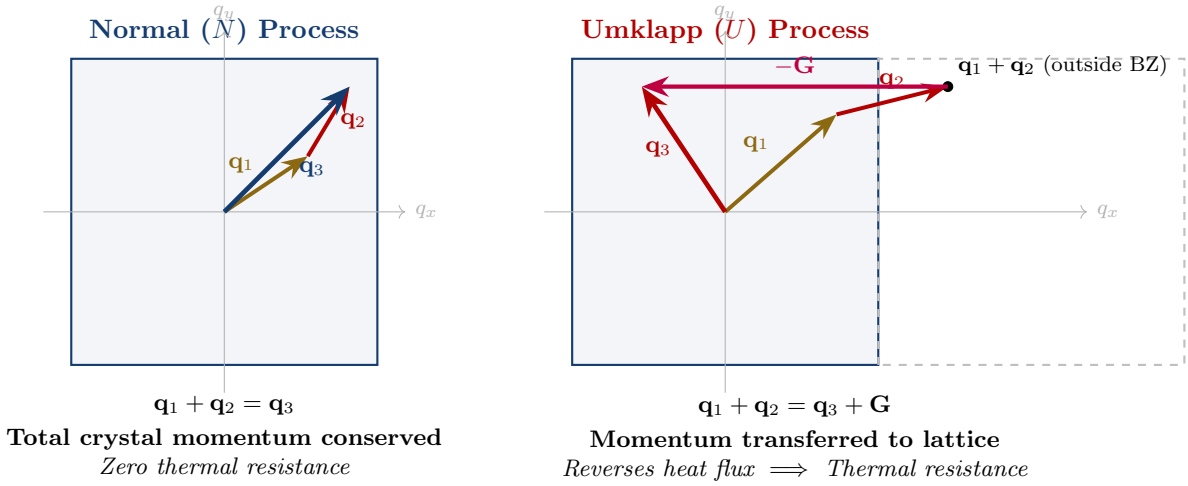


Figure 18: Comparison of three-phonon scattering geometries in reciprocal space. (Left) Normal (N) process: sum vector $\mathbf{q}_3$ remains inside the first Brillouin zone; total phonon momentum is conserved and heat continues forward unimpeded. (Right) Umklapp (U) process: the sum lands outside the first Brillouin zone and is folded back by reciprocal lattice vector $\mathbf{G}$, flipping the net phonon group velocity and creating finite thermal resistance.

8.4 Grüneisen Parameter and Equation of State (7 min)

Mode Grüneisen parameter:

$$\gamma_{\mathbf{k}s} = -\frac{V}{\omega_{\mathbf{k}s}} \frac{\partial \omega_{\mathbf{k}s}}{\partial V}$$

Average Grüneisen parameter:

$$\gamma = \frac{\sum_{\mathbf{k}s} \gamma_{\mathbf{k}s} C_{\mathbf{k}s}}{\sum_{\mathbf{k}s} C_{\mathbf{k}s}}$$

Mie–Grüneisen Equation of State

$$P = -\frac{dU_0}{dV} + \gamma \frac{U_{vib}}{V}$$

8.5 Computational Illustration (3 min)

Python: thermal expansion from an asymmetric potential

```python
import numpy as np
from scipy.integrate import quad

# Harmonic vs. anharmonic oscillator Boltzmann average <x>(T)
def U(x, C2=1.0, C3=0.15, C4=0.02):
    return 0.5*C2*x**2 + (1.0/6.0)*C3*x**3 + (1.0/24.0)*C4*x**4

def avg_x(T, anharm=True):
    kB = 1.0
```

```
    beta = 1.0/(kB*T)
    num, _ = quad(lambda x: x*np.exp(-beta*U(x)), -2.0, 3.0)
    den, _ = quad(lambda x:   np.exp(-beta*U(x)), -2.0, 3.0)
    return num/den

Tvals = np.linspace(0.05, 2.0, 30)
avgs  = [avg_x(T) for T in Tvals]
print("Anharmonic <x>(T)  ->  ", np.round(avgs, 4))
# For the harmonic potential, <x>(T) = 0 by symmetry (try setting C3=C4=0).
```

Tutorial / Practice Problems

1. Explain why negative thermal expansion occurs in some materials (e.g. ZrW_2O_8).
2. Calculate the Grüneisen parameter for a monoatomic chain if $\omega = \omega_0(a/r)^n$.
3. Why does thermal conductivity increase initially with T at very low T before decreasing?

9 Lecture 16: Tutorial & Problem Solving – Lattice Vibrations

Duration: 50 minutes

Problem Set

Problem 1

A monoatomic linear chain has $M = 40$ amu, $a = 3$ Å, and $C = 50$ N/m.

(a) Calculate the maximum phonon frequency.

(b) Calculate the sound velocity.

(c) At what wavevector does the group velocity equal half the sound velocity?

Worked Solution

(a) $\omega_{\max} = 2\sqrt{C/M} = 2\sqrt{\dfrac{50}{40 \times 1.66 \times 10^{-27}}} = 2.74 \times 10^{13}$ rad/s

(b) $v_s = a\sqrt{C/M} = 3 \times 10^{-10} \times \sqrt{\dfrac{50}{40 \times 1.66 \times 10^{-27}}} = 4.12 \times 10^{3}$ m/s

(c) $v_g = v_s \cos(ka/2) = v_s/2 \;\Rightarrow\; \cos(ka/2) = \tfrac{1}{2} \;\Rightarrow\; ka/2 = \pi/3 \;\Rightarrow\; k = \dfrac{2\pi}{3a}$

Problem 2

For a diatomic chain with $M = 2m$, show that the frequency gap at the zone boundary is

$$\Delta\omega = \sqrt{2C/m} - \sqrt{C/M}.$$

Worked Solution

Step 1: Zone boundary frequencies ($k = \pi/a$). At the first Brillouin-zone boundary $ka/2 = \pi/2$, so $\sin^2(ka/2) = 1$. Substituting into the general dispersion relation:

$$\omega_{\pm}^2 = C\left(\frac{1}{M} + \frac{1}{m}\right) \pm C\sqrt{\left(\frac{1}{M} + \frac{1}{m}\right)^2 - \frac{4}{Mm}}$$

Recognising that $\left(\frac{1}{m} - \frac{1}{M}\right)^2 = \left(\frac{1}{M} + \frac{1}{m}\right)^2 - \frac{4}{Mm}$:

$$\omega_{\pm}^2 = C\left(\frac{1}{M} + \frac{1}{m}\right) \pm C\left(\frac{1}{m} - \frac{1}{M}\right)$$

Since $M > m \implies \frac{1}{m} > \frac{1}{M}$:

$$\omega_+^2 = \frac{2C}{m} \implies \omega_+ = \sqrt{\frac{2C}{m}}, \qquad \omega_-^2 = \frac{2C}{M} \implies \omega_- = \sqrt{\frac{2C}{M}}$$

Step 2: Calculate the frequency gap for $M = 2m$.

$$\omega_- = \sqrt{\frac{2C}{2m}} = \sqrt{\frac{C}{m}}$$

Therefore, the forbidden frequency gap is:

$$\Delta\omega = \omega_+ - \omega_- = \sqrt{\frac{2C}{m}} - \sqrt{\frac{C}{m}} = \left(\sqrt{2} - 1\right)\sqrt{\frac{C}{m}} \approx 0.414\sqrt{\frac{C}{m}}$$

Problem 3

The Debye temperature of NaCl is 321 K. Calculate its specific heat at 30 K.

Worked Solution

Since $T = 30\ \text{K} \ll \Theta_D = 321\ \text{K}$:

$$C_V = \frac{12\pi^4}{5} R \left(\frac{T}{\Theta_D}\right)^3 = \frac{12\pi^4}{5} \times 8.314 \times \left(\frac{30}{321}\right)^3$$

$$C_V = 233.8 \times (0.0935)^3 = 0.191\ \text{J/mol}\cdot\text{K}$$

Problem 4

In a certain crystal, phonons have a linear dispersion $\omega = vk$ with $v = 3000$ m/s. The crystal has $N = 10^{23}$ atoms in volume $V = 10^{-3}$ m^3. Calculate:

(a) The Debye frequency
(b) The Debye temperature
(c) The specific heat at 10 K

Worked Solution

(a) Debye frequency ω_D: The total number of modes in volume V is $3N = \frac{V\omega_D^3}{2\pi^2 v^3}$, so

$$\omega_D = v\left(6\pi^2 \frac{N}{V}\right)^{1/3}$$

Given $N/V = 10^{23}/10^{-3} = 10^{26}$ m^{-3} and $v = 3000$ m/s:

$$6\pi^2 \times 10^{26} \approx 59.218 \times 10^{26} = 5.9218 \times 10^{27}\ \text{m}^{-3}$$

$$\left(5.9218 \times 10^{27}\right)^{1/3} \approx 1.809 \times 10^{9}\ \text{m}^{-1}$$

$$\omega_D = 3000 \times 1.809 \times 10^{9} = 5.43 \times 10^{12}\ \text{rad/s}$$

(b) Debye temperature Θ_D:

$$\Theta_D = \frac{\hbar\omega_D}{k_B} = \frac{1.0546 \times 10^{-34}\ \text{J}\cdot\text{s} \times 5.43 \times 10^{12}\ \text{s}^{-1}}{1.3807 \times 10^{-23}\ \text{J/K}} = 41.5\ \text{K}$$

(c) Specific heat at $T = 10$ K: Here $T/\Theta_D = 10/41.5 = 0.241$. Applying the Debye T^3 law:

$$C_V = \frac{12\pi^4}{5} N k_B \left(\frac{T}{\Theta_D}\right)^3 = \frac{12\pi^4}{5} \times (10^{23} \times 1.3807 \times 10^{-23}) \times (0.241)^3$$

$$C_V \approx 233.78 \times 1.3807 \times 0.0140 \approx 4.52\ \text{J/K}$$

Problem 5

The thermal conductivity of a dielectric crystal is 100 W/m·K at 300 K and 10 W/m·K at 100 K. Assuming the high-T behaviour $\kappa \propto 1/T$, verify consistency and estimate the mean free path at 300 K given $C_V = 2.5 \times 10^6$ J/m^3·K and $v_s = 4000$ m/s.

Worked Solution

Step 1: Check high-T consistency. For Umklapp-dominated phonon transport at high temperature, $\kappa \propto 1/T$, which implies $\kappa \cdot T = \text{constant}$.

$$\kappa(300\text{ K}) \times 300 = 100 \times 300 = 30000\text{ W/m}$$

$$\kappa(100\text{ K}) \times 100 = 10 \times 100 = 1000\text{ W/m} \neq 30000\text{ W/m}$$

Since $\kappa(100\text{ K}) < \kappa(300\text{ K})$, at 100 K the crystal is in the low-temperature boundary-limited regime ($C_V \propto T^3$ dominance) rather than the Umklapp regime.

Step 2: Estimate phonon mean free path ℓ at 300 K. From the kinetic kinetic formula for thermal conductivity:

$$\kappa = \frac{1}{3} C_V v_s \ell \implies \ell = \frac{3\kappa}{C_V v_s}$$

Substituting $\kappa = 100\text{ W/m}\cdot\text{K}$, $C_V = 2.5 \times 10^6\text{ J/m}^3\cdot\text{K}$, and $v_s = 4000\text{ m/s}$:

$$\ell = \frac{3 \times 100}{2.5 \times 10^6 \times 4000} = \frac{300}{1.0 \times 10^{10}} = 3.0 \times 10^{-8}\text{ m} = 30\text{ nm}$$

Computational Challenge

Simulate the time evolution of a Gaussian wave packet on a monoatomic chain (e.g. using velocity-Verlet integration of the coupled equations of motion from Lecture 8) and observe its dispersion – i.e. the spreading and reshaping of the packet as it propagates, a direct dynamical consequence of the nonlinear $\omega(k)$ relation derived at the start of this module.

Module 2 Summary Sheet

Dispersion Relations

- Monoatomic chain: $\omega(k) = 2\sqrt{C/M}\,|\sin(ka/2)|$
- Diatomic chain: $\omega_{\pm}^2 = C\left(\frac{1}{M}+\frac{1}{m}\right) \pm C\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4\sin^2(ka/2)}{Mm}}$

Density of States

- 1D: $g(\omega) = \dfrac{2L}{\pi\sqrt{\omega_{\max}^2 - \omega^2}}$
- d-dimensional (linear, isotropic dispersion): $g(\omega) \propto \omega^{d-1}$

Specific Heat Models

- Classical: $C_V = 3Nk_B$ (constant – fails at low T)
- Einstein: $C_V = 3Nk_B(\Theta_E/T)^2\, e^{\Theta_E/T}/(e^{\Theta_E/T}-1)^2$
- Debye: $C_V \to 3Nk_B$ (high T), $C_V = \dfrac{12\pi^4}{5}Nk_B(T/\Theta_D)^3$ (low T)

Thermal Conductivity

$\kappa = \dfrac{1}{3}C_V v_s \ell$, $\quad \kappa \propto T^3$ (low T, boundary-limited), $\quad \kappa \propto 1/T$ (high T, Umklapp-limited)

Looking ahead

The density-of-states and Bose-statistics machinery built here reappears in Module 3 for the *free electron gas* – except electrons are fermions, so n_B is replaced by the Fermi–Dirac function. Keep this parallel in mind: it is the fastest way to absorb the electronic specific heat result in the next module.

Further Reading

References

[1] N. W. Ashcroft and N. D. Mermin, *Solid State Physics* (Saunders, 1976). Chs. 4–5 (lattice vibrations, thermal properties). The standard graduate reference; the 1D chain derivations in this module follow Ch. 4 closely.

[2] C. Kittel, *Introduction to Solid State Physics*, 9th ed. (Wiley, 2021). Chs. 4–5. A more pedagogical alternative; well-suited for self-study.

[3] J. M. Ziman, *Elements of Advanced Quantum Theory* (CUP, 1969) / *Principles of the Theory of Solids*, 2nd ed. (CUP, 1972). Ziman is the place to read about Umklapp resistance and the Peierls argument with the original physical insight.

[4] G. P. Srivastava, *The Physics of Phonons* (Adam Hilger, 1990). Comprehensive treatment of anharmonic effects, with explicit derivations of the Grüneisen parameter and the Boltzmann transport equation for phonons.

[5] G. D. Mahan, *Many-Particle Physics*, 3rd ed. (Springer, 2000). The second-quantised treatment of phonons (Ch. 1) and a very clean derivation of the Debye specific heat (Ch. 3).

[6] A. Togo and I. Tanaka, "First principles phonon calculations in materials science," *Scripta Materialia* **108**, 1–5 (2015). The reference for the modern open-source `phonopy` package; useful for students who want to compute phonon dispersions and DOS for real materials.

[7] B. N. Brockhouse, "Phonons and the properties of matter," in *Inelastic Scattering of Neutrons in Solids and Liquids*, IAEA (1963). A historical but still very readable account of the experimental side.

[8] N. F. Ramsey, *Thermometry below 1 K: a continuing challenge*, in *Experimental Methods in the Physical Sciences* (Academic, 1992). An experimentalist's perspective on low-T specific-heat measurements.